CLINICAL · LARYNX & AIRWAY

Vocal fold paralysis

A paralysed vocal fold is a sign that something has happened along a very long nerve: in the brainstem, at the skull base, in the neck or deep in the chest. One fold paralysed costs the voice and sometimes the swallow; both paralysed near the midline cost the airway while sparing the voice. Find the cause, especially the lung or thyroid cancer that may be hiding behind a hoarse voice, and then choose between voice, airway and safe swallowing.

AT A GLANCE

- Unilateral affects voice; bilateral affects airway.
- Image the whole nerve path.
- Voice therapy, injection or medialisation.
- Secure the airway in bilateral palsy.

1 CHAPTER ONE Nerve supply

Two nerves, one long way round.

1.1 Motor

Every muscle that moves the vocal fold is supplied by the *recurrent laryngeal nerve*, except the *cricothyroid* (a tensor), which is supplied by the *external laryngeal nerve*, a branch of the superior laryngeal nerve. The interarytenoid receives fibres from both sides.

1.2 Sensory

Above the vocal folds, the internal laryngeal nerve; below them, the recurrent laryngeal nerve.

1.3 Course

- **Right recurrent nerve:** leaves the vagus at the root of the neck, hooks under the subclavian artery and climbs in the tracheo-oesophageal groove.
- **Left recurrent nerve:** leaves the vagus in the mediastinum, hooks under the arch of the aorta beside the ligamentum arteriosum, and climbs a much longer course; hence it is paralysed far more often.
- **Superior laryngeal nerve:** from the inferior ganglion of the vagus, behind the internal carotid artery, dividing at the greater horn of the hyoid into the internal (sensory, through the thyrohyoid membrane) and external (to cricothyroid, beside the superior thyroid artery) branches.

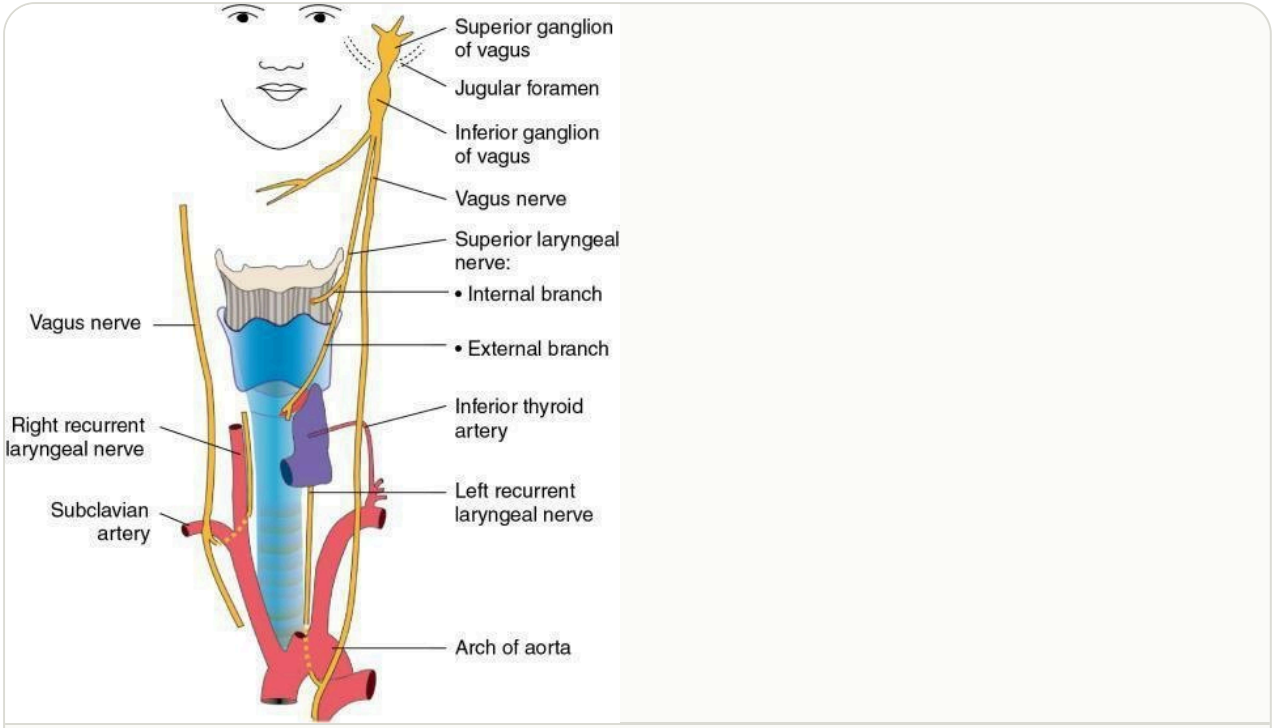
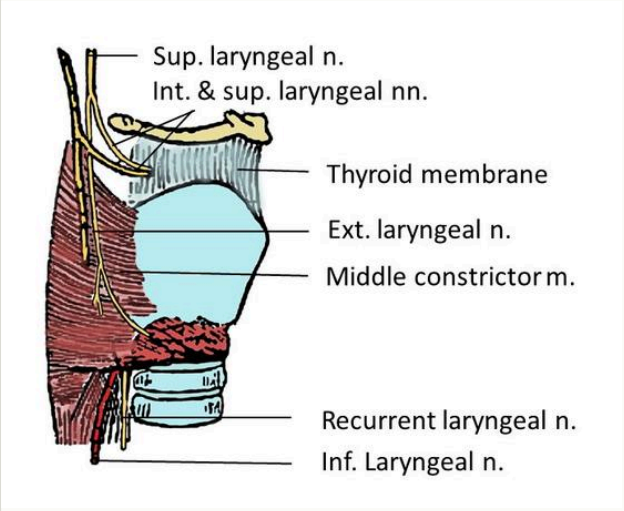


FIGURE 1 The recurrent and superior laryngeal nerves and their course: the left recurrent nerve loops around the arch of the aorta, the right around the subclavian artery.

Dhingra, fig. 60.1



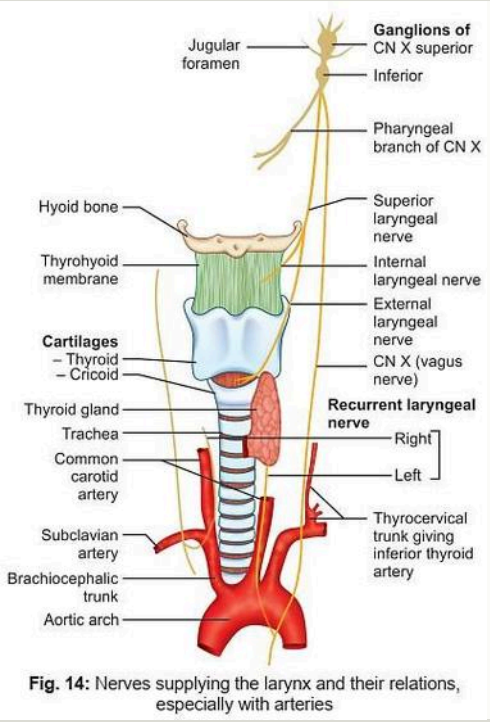


FIGURE 2 The nerve supply of the larynx, and the nerves with their relations.

Anatomy atlas · Doss p. 175, ENT Steps p. 251

2 CHAPTER TWO
Positions of the cord

What the folds show.

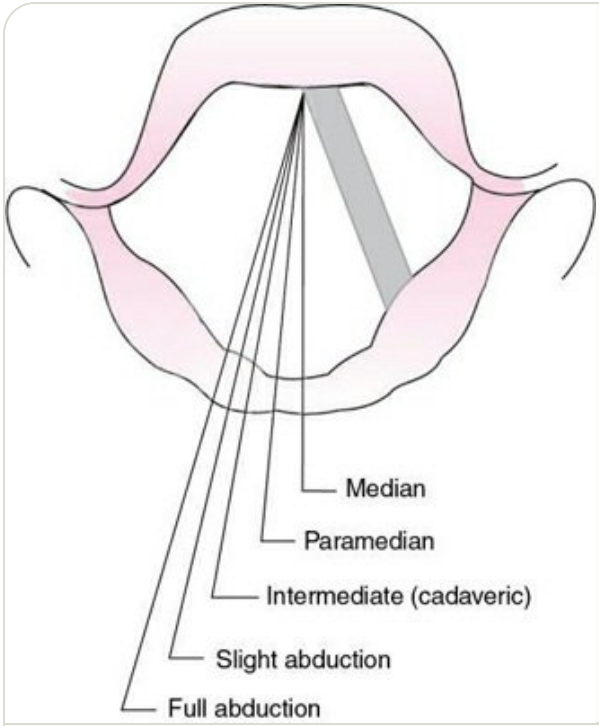


FIGURE 3 Positions of the vocal folds: median, paramedian, intermediate (cadaveric), slight abduction and full abduction.
Dhingra, fig. 60.2

POSITIONS OF THE VOCAL FOLD

POSITION	DISTANCE FROM MIDLINE	IN HEALTH	IN DISEASE
Median	0	Phonation	Recurrent nerve paralysis
Paramedian	1.5 mm	Strong whisper	Recurrent nerve paralysis
Intermediate (cadaveric)	3.5 mm	Neutral resting position	Combined recurrent and superior nerve paralysis
Gentle abduction	7 mm	Quiet breathing	Paralysis of adductors
Full abduction	9.5 mm	Deep inspiration	None

2.1 Why the paralysed cord lies near the midline

- **Semon's law:** in progressive organic lesions the abductor fibres, phylogenetically newer, are affected before the adductors, so the fold first lies near the midline.
- **Wagner–Grossman hypothesis:** with the recurrent nerve cut and the superior nerve intact, the cricothyroid still adducts the fold slightly into the paramedian position; with both nerves lost the fold lies in the cadaveric position.

Lee adds that the position also reflects partial or misdirected *regeneration*, which restores tone (and synkinesis) without useful movement; a fold often lies laterally at first and medialises over months.

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3 CHAPTER THREE Causes

Along the whole course.

3.1 By level

- **Supranuclear:** rare (bilateral cortical lesions).
- **Nuclear (nucleus ambiguus, medulla):** stroke (lateral medullary syndrome), tumours, motor neurone disease, polio, syringobulbia, multiple sclerosis, Arnold–Chiari malformation; other cranial nerves are involved.
- **High vagal (with superior laryngeal involvement, a combined paralysis):** posterior fossa tumours and basal meningitis; skull base fractures, nasopharyngeal carcinoma, glomus tumours; parapharyngeal tumours, metastatic nodes, lymphoma and penetrating injury.
- **Low vagal and recurrent nerve:** see the table.
- **Systemic:** diabetes, viral neuritis, diphtheria, syphilis, lead poisoning.
- **Idiopathic:** about 20 to 30%, many presumed viral; a diagnosis of exclusion.

CAUSES OF RECURRENT LARYNGEAL NERVE PARALYSIS

SITE	RIGHT	LEFT	BOTH
Neck	Thyroid surgery, thyroid cancer, cervical oesophageal cancer, nodes, trauma, subclavian aneurysm	The same	Thyroidectomy, thyroid cancer, cervical oesophageal cancer
Chest	Carcinoma of the right lung apex, tuberculosis of the pleura	Bronchogenic carcinoma, mediastinal nodes, thoracic oesophageal cancer, aortic aneurysm, enlarged left atrium (Ortner syndrome), cardiac and thoracic surgery, ductus ligation	Rare

LEFT VOCAL FOLD PALSY

Bronchogenic carcinoma must be excluded in every unexplained left-sided paralysis: image the whole course of the nerve from the skull base to the aortic arch with contrast CT (or MRI and CT), and examine the thyroid, oesophagus and neck.

3.2 Commonest causes overall

Malignancy (lung, thyroid, oesophagus) and surgery (thyroid, cervical spine through an anterior approach, cardiothoracic) lead the list; thyroidectomy is the commonest cause of bilateral paralysis.

4 CHAPTER FOUR Recurrent nerve paralysis

One or both.

4.1 Unilateral

All intrinsic muscles on that side are paralysed except cricothyroid, and the fold lies median or paramedian. About a third of patients have no symptoms; others have a weak, breathy voice that tires, diplophonia, loss of projection and range, and

sometimes aspiration of thin liquids. The voice often improves over months as the healthy fold learns to cross the midline.

4.2 Bilateral (bilateral abductor paralysis)

Usually after thyroidectomy, or neuritis. Both folds lie median or paramedian, held by the unopposed cricothyroids: *the voice is good but the airway is poor*, with stridor and dyspnoea that worsen on exertion and with any respiratory infection, sometimes requiring an emergency tracheostomy.

> **THE PARADOX**
>
> Bilateral recurrent palsy gives a good voice and a bad airway; bilateral combined palsy gives a bad voice, no airway problem and dangerous aspiration.

5 CHAPTER FIVE Superior laryngeal and combined paralysis

Tension and sensation.

5.1 Superior laryngeal nerve paralysis

Paralysis of the cricothyroid with numbness of the supraglottis on that side; usually part of a combined lesion, or the external branch injured in thyroid surgery. The voice is weak and cannot be raised in pitch, which singers notice at once. The glottis looks *askew*, with the anterior commissure rotated to the healthy side; the paralysed fold is shorter, lax and wavy, and flaps (sagging on inspiration, bulging on expiration). Electromyography of the cricothyroid confirms it. Voice therapy.

- **Bilateral:** paralysis with anaesthesia of the upper larynx causes repeated aspiration, cough and choking, and a weak, husky voice; it may recover after neuritis; tracheostomy with a cuffed tube and tube feeding, and epiglottopexy for persistent aspiration; cricothyroid approximation raises the pitch.

5.2 Combined (complete) paralysis

- **Unilateral:** after thyroidectomy injuring both nerves, or with lesions of the nucleus ambiguus or vagus above the origin of the superior laryngeal nerve. The fold lies in the *cadaveric position*, the healthy fold cannot meet it, and the result is a hoarse, breathy voice, aspiration of liquids and a weak cough.
- **Bilateral:** both folds cadaveric with an insensitive larynx: aphonia, aspiration, inability to cough and recurrent pneumonia. Tracheostomy and gastrostomy, then procedures that protect the lungs: epiglottopexy, vocal fold plication (both reversible), laryngotracheal separation or diversion, and, as a last resort, laryngectomy.

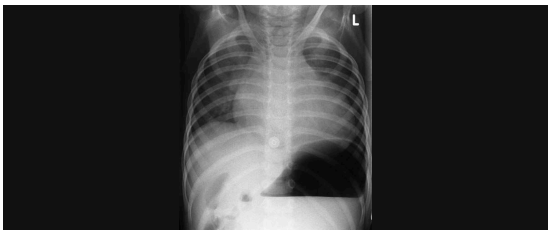
6 CHAPTER SIX Evaluation

Paralysis or fixation?

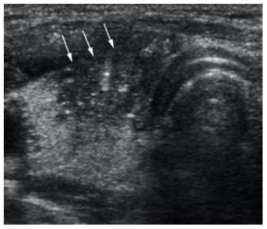
- **Laryngoscopy:** flexible and stroboscopic, observing movement, position, the glottic gap, the level of the folds and arytenoid tilt; confirm with phonation, sniffing and coughing.
- **Fixation versus paralysis:** posterior glottic scarring, cricoarytenoid arthritis or dislocation and tumour can fix a fold; at direct laryngoscopy the arytenoid is *palpated* for passive mobility.
- **Laryngeal electromyography:** distinguishes paralysis from fixation, shows synkinesis and helps predict recovery (absent or synkinetic activity predicts poor recovery).
- **Imaging:** contrast CT from the skull base to the aortic arch, or MRI of the brainstem and skull base for high lesions; chest imaging; thyroid ultrasound.
- **Swallowing assessment:** FEES or videofluoroscopy for aspiration.
- **Blood tests** where indicated (glucose, serology).

IMAGING IN THE APP 2 cases

Recurrent laryngeal nerve paralysis and thyroid cancer on imaging.



Paralysis of the Recurrent Laryngeal Nerve
Infection & inflammation · 9 images



Thyroid cancer
Tumours · 10 images

7 CHAPTER SEVEN Treatment

Voice, airway, swallow.

7.1 Unilateral paralysis

The aim is to close the glottic gap.

- **Voice therapy** early, while awaiting recovery or compensation over six to twelve months.
- **Injection laryngoplasty** (medialisation by injection, through the mouth or the neck, awake or under anaesthesia): temporary fillers (carboxymethylcellulose 2 to 3 months, hyaluronic acid 4 to 9 months, collagen) early, even within weeks, especially if aspiration is a problem; longer-lasting calcium hydroxylapatite (about 18 months) and autologous fat (variable survival). Teflon is abandoned because of granulomas.
- **Type I thyroplasty (medialisation laryngoplasty):** a window in the thyroid lamina and a silastic or Gore-Tex implant in the paraglottic space push the fold medially; done under local anaesthesia so that the voice can be tuned; for permanent paralysis with a gap of a few millimetres and symmetrical arytenoids.
- **Arytenoid adduction:** a suture through the muscular process rotates the arytenoid inwards, mimicking the lateral cricoarytenoid; for a wide posterior gap, a lateralised vocal process or a level difference; usually with thyroplasty.
- **Reinnervation:** ansa cervicalis to recurrent nerve anastomosis restores bulk and tone (not movement); useful when the nerve is known to be cut and in children, who would outgrow an implant.

7.2 Bilateral abductor paralysis

The aim is an airway without ruining the voice; any procedure that widens the glottis worsens the voice.

- **Tracheostomy:** the safest, the gold standard; the voice is preserved with finger occlusion or a speaking valve.
- **Endoscopic glottic widening:** *posterior transverse cordotomy* (Kashima), cutting the fold off the vocal process with the laser; *partial medial arytenoidectomy*; suture *lateralisation* of the fold; repeated if needed.
- Older external procedures: arytenoidectomy and arytenoidopexy (Woodman), and type II thyroplasty.
- Botulinum toxin to the adductors (temporary); reinnervation and laryngeal pacing (selected, experimental).
- In the first months after thyroidectomy, a temporary tracheostomy and waiting is wise, since many palsies recover.

OPERATIVE GUIDES IN THEATRE 5 operations

Injection, medialisation, intraoperative medialisation, bilateral immobility and protection of the nerve.

Vocal Fold Injection via Microlaryngoscopy
Injection

Medialization Laryngoplasty and Arytenoid Adduction
Medialisation

Intraoperative Medialization Laryngoplasty
Intraoperative medialisation

Bilateral Vocal Fold Immobility
Bilateral immobility

Open Thyroidectomy
Protecting the nerve

7.3 Isshiki's thyroplasties

- **Type I:** medialisation.
- **Type II:** lateralisation, for the airway.
- **Type III:** shortening and relaxation, to lower the pitch (mutational falsetto, female-to-male voice).

- **Type IV:** lengthening and tensing, to raise the pitch (male-to-female voice, bowed or lax folds).

8 CHAPTER EIGHT Congenital paralysis

In infants.

Unilateral (birth trauma, cardiac and great vessel anomalies, cardiac surgery) gives a weak cry and feeding problems; bilateral (Arnold–Chiari malformation, hydrocephalus, birth haemorrhage, meningocele) gives stridor needing tracheostomy. Many recover; definitive surgery is delayed (see congenital lesions of the larynx).

9 CHAPTER NINE For the examination

What the examiner wants to hear.

9.1 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- **Nerve supply** and why the left recurrent nerve is more often paralysed.
- **Positions** of the cord, Semon's law and the Wagner–Grossman hypothesis.
- **Causes** by level; exclude carcinoma of the lung.
- **Unilateral versus bilateral** recurrent palsy: voice and airway.
- **Superior laryngeal nerve palsy:** askew glottis, loss of pitch.
- **Treatment:** injection, thyroplasty and its types, arytenoid adduction, cordotomy and tracheostomy.

9.2 Common pitfalls

COMMON PITFALLS

- Calling a left palsy idiopathic without imaging to the aortic arch.

- Treating a fixed arytenoid as a paralysis without palpation or electromyography.
- Doing an irreversible glottic widening soon after thyroidectomy when the nerve may recover.

Written for this app from Oxford Textbook of Otolaryngology ch. 47, 71, Oxford Specialist Handbook ch. 8, Oxford Handbook of ENT ch. 12, Dhingra ch. 60 and K. J. Lee ch. 50. Figures reproduced from the books for private study. Study reference only: follow local protocols and senior advice for patients.

IMAGING

On imaging

The imaging of this topic: key findings, differentials and representative scans.

LARYNX & TRACHEA · INFECTION & INFLAMMATION

Paralysis of the Recurrent Laryngeal Nerve

IMAGING FINDINGS

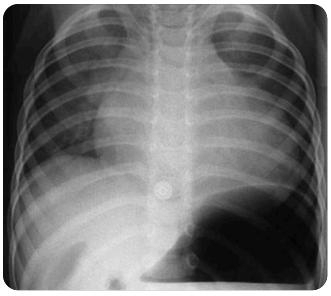
- Medially rotated arytenoid and dilated ventricle
- Dilated pyriform sinus; medial, thickened aryepiglottic fold
- Atrophy of the posterior cricoarytenoid and thyroarytenoid

TOP DIFFERENTIALS

- Arytenoid dislocation after trauma
- Glottic carcinoma
- Laryngocele

CHECKLIST

- Scan from skull base to the aortopulmonary window on the left
- Thyroid, oesophagus and mediastinal nodes



A 7-year-old child with hoarseness and left vocal cord paralysis. Chest radiograph, posteroanterior view. The grossly dilated heart is compressing…



CECT in a patient with new-onset hoarseness shows slightly medially rotated left arytenoid cartilage & a more prominent dilated left laryngeal…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 296.

THYROID & PARATHYROID · TUMOURS

Thyroid cancer

IMAGING FINDINGS

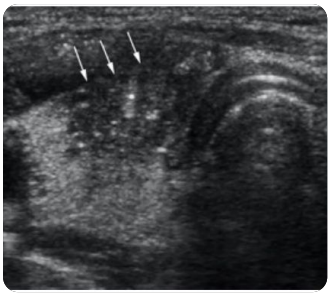
- Solid hypoechoic nodule with microcalcifications, taller than wide
- Irregular margins or extrathyroidal extension
- Cystic or calcified nodes in levels III, IV and VI

TOP DIFFERENTIALS

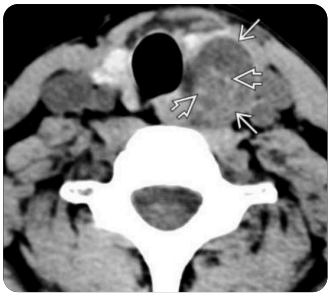
- Benign adenoma or colloid nodule
- Multinodular goitre
- Medullary or anaplastic carcinoma

CHECKLIST

- Trachea, oesophagus, larynx and recurrent nerve
- Iodinated contrast delays radioactive iodine



16-year-old lady presenting with hard right neck lump confirmed to be papillary ca on cytology. US shows strongly hypoechoic lesion with…



Axial NECT shows a well-defined mass arising from the left thyroid lobe with fine, speckled microcalcifications centrally. Small calcifications such as…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 238.